1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that immune contexture (Immunoscore/TILs), genomic instability (MSI/dMMR), and dynamic tumor burden (ctDNA) each predict immunogenicity, yet some MIBC tumors with apparently favorable immune contexture (high TILs) or high mutational load still fail to derive durable benefit from immune checkpoint blockade; why are immune-infiltrated tumors sometimes non■responsive?

Working Hypothesis

In muscle-invasive bladder cancer (MIBC), the predictive value of pre-treatment tumor-infiltrating lymphocyte (TIL) density for response to immune checkpoint blockade is contingent on mismatch repair (MMR) status defined by MLH1 and MSH2. Specifically, MLH1/MSH2 loss-of-function (MMR-deficient) tumors with high TIL density will show superior immunotherapy response (higher objective response rate and longer duration of response) compared with MLH1/MSH2-intact tumors, where high TIL density will have limited predictive value due to predominance of bystander or exhausted T cells. A combined biomarker model incorporating TIL density and MLH1/MSH2 LOF will outperform TIL density alone in discriminating responders from non-responders (therapy_response). Secondary modifiers—TP53 mutation status, GARS1 expression, urine DNA methylation panels, and HPV status—will modulate the strength of the TIL/MMR interaction and further refine misclassification, providing a more robust, multilevel framework for predicting immunotherapy outcomes in MIBC.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MLH1, MSH2
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MLH1: MLH1/MSH2 (max 2) to comply with the Focusing Rule and placed ctDNA, TP53, GARS1, urine methylation panels, and HPV-status into secondary_modifiers for follow-up analyses.

MSH2: MSH2 (max 2) to comply with the Focusing Rule and placed ctDNA, TP53, GARS1, urine methylation panels, and HPV-status into secondary_modifiers for follow-up analyses.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	ctDNA (liquid biopsy dynamics)	Supporting modifier from knowledge graph

Secondary	TP53	Supporting modifier from knowledge graph
Secondary	GARS1	Supporting modifier from knowledge graph
Secondary	urine DNA methylation biomarker panels	Supporting modifier from knowledge graph
Secondary	HPV status	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: elevates neoantigen load and enhances T cell priming, enabling high-TIL tumors to mount productive anti-tumor responses. However, in MMR-proficient tumors, high TIL density often reflects bystander or exhausted T cells, yielding poorer responses.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- ctDNA (liquid biopsy dynamics): Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TP53: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.